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SUPERSEDING
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MILITARY SPECIFICATION

BATTERY, STORAGE: SILVER-ZINC, RECHARGEABLE; GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for silver-zinc rechargeable batteries and cells.

1.2 Classification. Batteries shall be of the following types and classes as specified on the applicable military specification sheet (see 6.2.1):

TYPE I - Batteries containing only individual monoblocks (four cells in a molded plastic case).
TYPE II - Batteries containing one or more monoblocks mounted in an external battery box.

CLASS 1 - 1 Monoblock
CLASS 2 - 2 Monoblocks
CLASS 3 - 3 Monoblocks
CLASS 4 - 4 Monoblocks
CLASS X - X Monoblocks, where X = number of monoblocks contained in battery box.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to Commandant of the Marine Corps, Headquarters Marine Corps (Code LMA-1), Washington, DC 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-S-571	- Solder; Tin Alloy, Lead-Tin Alloy and Lead Alloy
PPP-B-636	- Box, Shipping, Fiberboard
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-B-601	- Box, Wood, Cleated Plywood
PPP-T-60	- Tape, Packaging, Waterproof

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-T-152	- Treatment, Moisture - and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-F-14256	- Flux, Soldering, Liquid (Rosin Base)
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment
MIL-E-16400	- Electronics, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for
MIL-B-26701	- Bottle Screw Cap and Carboys, Polyethylene Plastic

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment
MIL-STD-45662	- Calibration Systems Requirements

2.1.2 Other Governement documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

MARINE CORPS

DL 5122-000-1100	- Battery, Storage, BB-451/U
DL 5122-000-1299	- Battery, Storage, BB-606/U

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connections with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard sample. A standard sample (see 6.5) may be provided for examination and use in assuring compliance with this specification.

3.2 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets and drawings cited thereon. Where there exists a conflict, the following order of precedence shall govern:

a. Contract
b. Specification sheet
c. This specification
d. Furnished standard sample
e. Drawings

3.3 First article. The first article is defined as a sample which will be identical to the batteries which the manufacturer will subsequently produce in the fulfillment of the contract. As soon as practicable after award of the contract, and prior to submission of any batteries for final acceptance, the contractor shall furnish 24 first article monoblocks for examination and test to determine conformance to this specification. Examination and testing shall be those specified herein. Approval of the first article inspection by the acquiring activity shall not relieve the contractor of his obligation to supply batteries conforming to this specification. Any changes or deviations of batteries from the first articles shall be subject to the approval of the contracting officer.

3.4 Parts and materials. Parts, materials and processes shall be as specified herein. The selection of class, grade, or type shall be such that, when mounted in the equipment, the part or material will perform its intended function under the service and storage conditions specified herein. The requirements of MIL-E-16400 shall be followed with the exception that the unacceptable materials list shall not apply to materials utilized within the monoblocks. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the

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work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended end use of the item. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials.

3.4.1 Plastic material and parts. Where not machined, plastic materials and parts shall have the original smooth or polished surfaces. Surfaces which have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.4.2 Securing of parts. Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts shall not occur when the items in which they are used are subjected to the shock tests specified herein. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. When practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such a manner that failure of a single rivet or screw shall not free the part completely.

3.4.3 Tropicalization. Tropicalization of the complete equipment is not required. Wherever practicable, materials used in the equipment shall not, in themselves, be nutrients for fungi or susceptible to deterioration by moisture and fungi. If not practicable to use such materials for certain items required in the equipment, such items shall be given moisture and fungus proofing treatment prior to assembly in the equipment or the areas in the equipment where such items are used shall be coated with a moisture and fungus resistant varnish in accordance with MIL-T-152 prior to shipment.

3.5 Design and construction. The batteries shall be designed and constructed as follows:

3.5.1 Monoblocks. The battery shall be made up of series-connected cells constructed in monoblocks composed of four cells per block which shall have met the performance requirements of this specification. The number and type of monoblocks required shall be as specified in the applicable specification sheet in accordance with the classification requirements of 1.2. The supporting grids used in the positive and negative plates shall be pure silver (99.9 percent). The buss tabs shall be pure silver (99.9 percent) and shall be installed so that they extend at least 2.50 inches (63.5 millimeters (mm)) below the top of the plates. Each monoblock shall contain a minimum of 16.2 troy ounces (503.88 grams) of silver (99.9 percent).

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3.5.2 Gassing. The average rate of emission of gas from each cell of a monoblock shall not exceed 3.3 milliliters (ml) per hour, per cell (corrected to standard temperature and pressure) when being tested in accordance with 4.10.4.10.

3.5.3 Battery box. The battery box for type II batteries shall be made of high strength material, shall have a smooth finish, and shall be free from pits, blow-holes, rough spots, or other deformations. It shall be designed and constructed in accordance with DL 5122-000-1100, DL 5122-000-1299 or specification sheet, as applicable, and the following:

(a) The monoblocks making up the complete battery shall be securely mounted to prevent movement within the battery box; however, the monoblocks and related terminal hardware shall be capable of being removed and replaced in both the dry and activated conditions by removing no more than 16 terminal nuts per monoblock.

(b) The terminals and vent plugs of the monoblock shall be accessible.

(c) The battery box shall be capable of maintaining its dimensions within specified tolerances during the lifetime of the battery, i.e., storage, testing, and operation.

3.5.4 Cover. The battery box cover for type II batteries shall be made of high strength material and shall have smooth sides except for holddown bars, latch hooks, etc. The dimensions of the cover, and the dimensions and locations of holddowns, latch hooks, carrying straps, etc., shall conform to DL 5122-000-1100, DL 5122-000-1299 or specification sheet, as applicable. The cover shall be sealed to the case with a gasket secured to the inside of the cover (or the case) and positioned so as to match the rim of the container body to form an effective fluid seal between the cover and the battery box body. In addition, the cover shall be permanently secured to the battery box with a nylon cord in such a manner that the cord is not visible when the cover is sealed to the battery box body.

3.5.5 Latches. The cover to type II batteries shall be secured to the battery box body. Latches, hooks or thumb screws shall be mounted as specified on DL 5122-000-1100, DL 5122-000-1299 or specification sheet, as applicable.

3.5.6 Terminals. Size of terminals, spacing and necessary hardware shall be as specified on DL 5122-000-1100, DL 5122-000-1299 or specification sheet, as applicable. They shall provide positive means of making and maintaining external electrical connections to other batteries or equipment. Contact surfaces shall be free of insulating compounds, corrosive salts or other contaminants. If screws, nuts, and other threaded parts are specified on the applicable drawings or specification sheet, they shall be American National

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coarse or fine thread in accordance with FED-STD-H28. Terminals shall be located as shown on the applicable specification sheet or drawing.

3.5.7 Intercell connectors. Unless otherwise specified herein, intercell connectors and other current carrying parts shall be fabricated of materials shown on drawings and shall have sufficient cross sectional area to withstand any charge or discharge rate specified in the specification sheet. They shall be removable and not damaged by the electrolyte. The dimensions shall be in accordance with DL 5122-000-1100, DL 5122-000-1299 or specification sheet, as applicable.

3.5.8 Battery filler kit. The battery filler kit shall be as follows:

(a) The kit shall include 4 individual containers of electrolyte. Each bottle shall contain a premeasured quantity of electrolyte to properly fill one cell of the battery. The electrolyte container shall conform to MIL-B-26701 and be fabricated of an alkali-resistant, nonbreakable plastic material which shall maintain the electrolyte in a usable condition, and be leakproof, in storage and shipment. Bottle closure requirements of MIL-B-26701 shall apply. Containers with screw cap closures must be sealed with a secondary closure. The secondary closure must be made using a heat sealing process to ensure an air tight seal over the opening of the bottle.

(b) The battery filler kit shall include items required for battery activation and battery parts in quantity as may be required for maintenance and replacement during the normal life of the battery. The battery operating instructions and, if required by the specification sheet, the charging log (service record form) shall also be provided with the kit.

(c) Filling nozzles included in the filling kit shall provide a leakproof connection both to the battery vent hole and to the electrolyte bottle during the activation process. No special tools or procedures other than normal hand tightening shall be required to obtain a leakproof configuration during normal handling for activation.

3.5.9 Wiring. Wiring shall be neat and sturdy and meet the following requirements.

3.5.9.1 Slack. Wire and cables shall be as short as practical except that sufficient slack shall be provided for the following:

(a) To provide for at least two replacements of the part to which the wire or cable is connected.

(b) To ensure freedom of motion of lugs or terminals normally intended to have some degree of movement.

(c) To facilitate field repair of broken or cut wires.

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3.5.9.2 Routing. Wires shall be routed and secured in a manner to avoid contact, under service conditions specified herein, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.5.9.3 Splicing. Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.5.10 Riveting. Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. The distance from the centers of rivet holes to the edges of the material in which the holes are formed shall be at least one and one-half times the rivet diameter. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.5.11 Soldering. Solder shall conform to QQ-S-571 with the exception of solder used within the monoblocks. For soldering external monoblock electrical connections, composition SN60, type AR or S, shall be used for general purpose and composition SN62 and SN63, type AR or S, for special purposes. When type S solder is used, the flux shall conform to MIL-F-14256.

3.5.11.1 Acid or acid salts. No acid or acid salts shall be used in preparation for, or during, soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning of soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.5.11.2 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material which has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.5.11.3 Protective coating. Solder joints used for external monoblock wiring connections shall be completely protected with a coating of room temperature cure epoxy.

3.5.12 Electrical requirements. The electrical requirements of the batteries shall be as follows.

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3.5.12.1 Insulation resistance. The insulation resistance between any two terminals not electrically connected and between all ungrounded terminals shall not be less than 5 megohms when tested in accordance with 4.10.4.1.

3.5.12.2 Discharge capacity. Batteries shall be capable of meeting the discharge capacity when tested in accordance with 4.10.4.5 and the requirements of the applicable specification sheet.

3.5.12.3 Electrical leakage. When the monoblock is tested in accordance with 4.10.4.2., the ohmmeter reading shall be not less than 100 megohms.

3.5.12.4 High rate discharge. Batteries shall be capable of meeting the high rate discharge capacity when tested in accordance with 4.10.4.6 and the requirements of the applicable specification sheet.

3.5.12.5 Retention of charge. Batteries shall be tested in accordance with 4.10.4.8 and shall meet the capacity specified on the applicable specification sheet.

3.5.12.6 Cycle life. Batteries shall meet the cycle life requirements of 4.10.4.9 and the applicable specification sheet.

3.5.13 Vents. Air circulation vents shall be as follows.

3.5.13.1 Monoblock vent plugs. Monoblocks, when provided with vent plugs, shall comply with the following:

(a) Prevent electrolyte leakage or spraying in any battery attitude during performance tests specified in 4.10.4 with exception that minute deposits of white carbonate are permissible.

(b) Maintain positive internal cell pressure and vent gasses prior to reaching 10 pounds force per square inch (lb_f/in^2) (68.95 kilo pascals (kPa), when tested in accordance with 4.10.1.1. Lower limit is 0 lbf/in^2 (0 kPa).

3.5.13.2 Battery box safety relief valve. The battery box safety relief valve used on type II batteries shall maintain positive internal pressure within the box and vent gasses prior to reaching 8 lb_f/in^2 (55.16 kpa) when tested in accordance with 4.10.1.2.

3.5.14 Internal pressure. Monoblocks tested in accordance with 4.10.2 shall not show a pressure drop of more than 0.5 lb_f/in^2 (3.45 kpa) over a 5-minute period.

3.5.15 Attitude. Batteries shall be designed for operation in any position without loss of capacity, voltage or electrolyte leakage during any of the tests specified herein.

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3.5.16 Final battery conditioning. Batteries shall be furnished completely assembled, with electrolyte kit (when required) in a charged and dry condition.

3.5.16.1 Rated ampere-hour capacity (1Cx). The batteries shall be capable of providing the rated ampere-hour capacity (1Cx) as required by the applicable specification sheet.

3.5.16.2 Voltage. Batteries shall be capable of supplying the maximum, nominal and final voltage as required on the applicable specification sheet.

3.5.17 Interchangeability. Like units, assemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment (see 4.6.1). Individual items shall not be handpicked for fit or performance.

3.6 Performance. The battery, tested for measured performance[1/] in accordance with tests specified herein, shall meet the specified performance[2/] requirements shown on the applicable specification sheet.

3.6.1 Environmental. Batteries, when subjected to the tests in Table I, shall not show the following:

(a) Dimensional distortion beyond limits specified on DL 5122-000-1100, DL 5122-000-1299 or specification sheet as applicable.

(b) Sharp current or voltage fluctuations during any specified discharge period, at any time during or following performance of each test.

(c) Mechanical failure of any part.

(d) Electrolyte leakage or spray of electrolyte.

(e) Diminution of the rated capacity specified in the applicable specification sheet during any specified discharge period at any time during or following the performance of each test.

1/ The actual numerical figure obtained in conducting a test in accordance with Section 4 of this specification, such as nominal voltage, final voltage, capacity, cycle life, etc., shall be known as the measured performance.

2/ The actual numerical figure given in the specification sheet as a performance requirement, such as nominal voltage, final voltage, capacity, cycle life, etc., shall be known as the specified performance.

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TABLE I. Environmental tests.

Tests	Test Paragraph
Mechanical shock	4.10.5.1
Altitude	4.10.5.2
Vibration	4.10.5.3

3.6.1.1 Immersion. Type II batteries shall show no evidence of leakage of water into the box after being tested in accordance with 4.10.5.4.

3.6.1.2 Temperature. The batteries shall withstand temperature requirements as follows:

(a) Operating - The battery in the wet, charged or discharged condition shall be capable of withstanding temperature exposure over the range specified on the specification sheet.

(b) Nonoperating - Batteries in a dry charged condition shall show no evidence of breaking or cracking of any component when subjected to the temperature range of -57°C to +65°C, when tested in accordance with 4.10.5.5, activated in accordance with 4.10.4.3 and discharged in accordance with 4.10.4.5.

3.6.1.3 Extreme temperature. When tested in accordance with 4.10.4.7, batteries shall be capable of providing the capacity specified on the specification sheet.

3.6.2 Chemical requirements. The chemical requirements of the batteries and electrolyte shall be as follows:

3.6.2.1 Corrosion resistance. External metal parts of the battery shall show no evidence of cracking, pitting, chipping, scaling, corrosion, or other deleterious effects after being tested in accordance with 4.10.3.1.

3.6.2.2 Alkali resistance of plastic materials. When tested in accordance with 4.10.3.2, plastic container materials shall show no cracks or blisters and shall not change more than 2.5 percent in weight or 2.0 percent in any dimension.

3.6.2.3 Alkali resistance of elastomer materials. All elastomer materials used in each type of battery assembly, after being tested in accordance with 4.10.3.3, shall cause no degradation of battery assembly performance. Pads, O-rings and gaskets shall have a permanent compression set of less than 6 percent after this test.

3.6.2.4 Electrolyte carbonates composition. The amount of potassium carbonates contained in the electrolyte solution shall not exceed 60 grams per liter when tested in accordance with 4.10.3.4.

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3.6.3 Activation and filled discharge. When activated with electrolyte and soaked in an upright position for a maximum of 24 hours, the battery shall be capable of meeting the requirements specified herein and shall meet the requirements of 4.10.4.3.

3.7 Dimension and weight. Dimensions and weight of the battery and monoblocks shall be as shown on DL 5122-000-1100, DL 5122-000-1299 or specification sheet, as applicable.

3.8 Battery box finish. Finishes used on type II battery boxes shall not scale, chip, or peel after the battery has been subjected to all the tests of this specification. Material used as a finish shall be resistant to 40 percent solution (1.400 ± 0.015 specific gravity) potassium hydroxide. Color shall be as specified on the contract (see 6.2).

3.9 Cleaning. The battery assemblies shall be cleaned as follows:

3.9.1 Parts. After fabrication, parts shall be cleaned in accordance with good commercial practice. Cleaning processes shall have no deleterious effect. All corrosive material shall be removed.

3.9.2 Battery and monoblock. After assembly, batteries and monoblocks shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly.

3.10 Identification and marking. Each battery and monoblock shall be provided with permanently mounted labels in accordance with the following requirements. Monoblocks may have labels hot-stamped on one side. Marking shall be in accordance with the requirements of MIL-P-15024. In addition, marking shall be insoluble in 40 percent solution (1.400 ± 0.015 specific gravity) potassium hydroxide.

3.10.1 Identification labels. Identification labels shall be in accordance with DL 5122-000-1100 or DL 5122-000-1299, as applicable and shall contain the following data:

Battery, Storage, Rechargeable BB-()3/

Zinc-Silver Oxide Alkaline
Nominal Voltage/Capacity/Rate/Temp
(i.e. 24V/25Ah/5hour/20oC)
NSN
Contract No
Date Code
Manufacturer's Name
Manufacturer's Plant Location

3/ The parentheses in the nomenclature will be deleted or replaced by a number or letter identifying the particular design as specified (see 6.2).

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3.10.2 Date code. The code shown shall indicate the month and year of manufacture of the battery by means of a four-digit number in which the first two digits shall indicate the month and the last two digits shall indicate the year. All months before the tenth month (October) shall be indicated by a single digit preceded by a "0". When a battery assembly is completed during the last three working days of a month or the first three working days of the subsequent month, the manufacturer is permitted to use either month as the coded month of manufacture.

3.10.3 Instruction labels. Each type II battery shall contain securely attached instruction labels. Labels shall provide instructions for operation and charging of the battery, first aid instructions, salvage instructions and a charging log. As a minimum, the charging log shall provide sufficient space (on one side, or on two sides if provision is made for flipping the log) for recording charging data for up to 50 charges. Instruction labels shall be in accordance with DL 5122-000-1100, DL 5122-000-1299 or specification sheet, as applicable.

3.10.4 Terminal markings. Terminal markings shall be indicated as follows:

3.10.4.1 Type I batteries. Portions of the positive terminals of type I batteries which are not part of the contact surface shall be painted red and portions of the negative terminal which are not part of the contact surface shall be painted black.

3.10.4.2 Type II batteries. Type II batteries shall have terminal markings in accordance with 3.10.4.1. In addition, type II batteries containing two separate external positive and negative terminals shall have the positive terminals legibly indicated by POS, P, or + and the negative terminals indicated by NEG, N, or -; the polarity markings shall be placed on the applicable terminal or as close thereto as possible. Depressed characters of sufficient size to be readily recognized shall be used.

3.10.5 Serial numbers. Unless otherwise specified in the contract or purchase order (see 6.2), each battery box cover and each monoblock shall be assigned a separate serial number. Serial numbers shall be located on the top of the battery box covers and on the top of the monoblocks (when possible) in such a manner that they are clearly legible without requiring removal of any components. The separate numbering sequences for the covers and monoblocks shall be established by the manufacturer.

3.11 Workmanship. The batteries shall be fabricated and finished of a quality prevailing among manufacturers normally producing similar batteries of the type specified herein. All parts of the internal assemblies shall be readily accessible for repair and replacement.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspections (see 4.5).
(c) Inspection of packaging (see 4.5.2).

4.3 First article inspection. When specified (see 6.2), first article inspection shall be performed by the contractor after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production.

4.3.1 First article inspection routine. The first article samples (see 4.3.2) shall be subjected to the inspection routine as specified in Table II in the order shown. Where Table II indicates "Battery Group Number," monoblocks shall be assembled in type II battery boxes as specified in 4.3.2.

4.3.2 First article sample. A sufficient quantity of type II battery boxes to contain 24 monoblocks shall be supplied by the contractor, where type II batteries are deliverable under the contract or purchase order, or as specified for type I battery acquisitions (see 6.2). Before testing, the resulting quantity of battery boxes shall be divided into three equal groups, and the resulting groups designated group 1, group 2, and group 3. Monoblocks shall be assigned numbers 1 through 24. After assignment of group numbers, batteries and their respective monoblocks shall not lose their assigned group number identity. The required number of sealed bottles of electrolyte shall be supplied by the contractor to properly fill each monoblock to operating condition. All unopened bottles of electrolyte shall be subjected to test specified for 4.10.6 in Test 1 before proceeding with the remaining first article tests. Failures attributed to battery box failure shall be cause for rejection only under acquisitions involving type II batteries.

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TABLE II. First article tests

Test Number	Test Paragraph	Test	Monoblock Number	Battery Group Number	Attitude
1.	4.10.6	Electrolyte container leak test			
	4.10.2	Internal pressure		1,2&3	upright
	4.10.5.5	Thermal shock		1 & 2	upright
	4.10.5.2	Altitude			
		Dry		1	upright
		Activated		2	side
	4.10.4.3	Filled discharge		1,2&3	upright
	4.6	Visual and mechanical 1/	1 thru 24	1,2&3	upright
	4.10	Vent	1 thru 16	1 & 2	upright
	4.10.4.1	Insulation resistance	1 thru 16		- - - -
	4.10.5.4	Immersion		1	upright
				2	inverted
2.	4.10.4.4	Charge		1 & 2	upright
	4.10.4.10	Gassing		1 & 2	upright
	4.10.4.5	Full capacity		1	side
		discharge		2	upright
3.	4.10.4.6	High rate discharge		1	upright
	4.10.4.5	Full capacity discharge		2	upright
4.	4.10.4.7	Extreme temperature		1	upright
	4.10.4.5	Full capacity discharge		2	upright
5.	4.10.4.4	Charge		1 & 2	upright
	4.10.5.1	Mechanical shock			
		Test I		1	bottom, side, face
		Test II		1 & 2	4 bottom corners
	4.10.5.3	Vibration		2	3 faces

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6.	4.10.5.4	Immersion[1/]	1 & 2	upright
	4.10.4.5	Full capacity discharge	1 & 2	upright
	4.10.4.5	Full capacity discharge	1	side
			2	upright
7.	4.10.4.8	Retention of charge		
		Standard temperature	1	upright
		High temperature	2	upright
8.	4.10.4.9	Cycle life	3	upright
9.	4.10.4.2	Leakage[1/]	1 thru 16	- -

1/ See 4.3.

4.3.3 First article materials inspection. First article inspection of materials shall consist of the tests specified in Table III. These tests shall be conducted on three samples of each material or component used in the battery. The samples shall be representative of the batteries furnished to the Government.

TABLE III. Chemical tests

Test	Req. para.	Insp. para.
Corrosion resistance	3.6.2.1	4.10.3.1
Alkali resistance (plastic)	3.6.2.2	4.10.3.2
Alkali resistance (elastomer)	3.6.2.3	4.10.3.3
Electrolyte Composition	3.6.2.4	4.10.3.4

4.4 Acceptance inspection. Acceptance inspection shall be performed on each unit supplied under the contract or purchase order. Acceptance inspection shall consist of visually inspecting each battery and its associated inspection records in order to determine compliance with 3.5.16.

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4.5 Quality conformance inspection. Quality conformance inspection shall be performed as follows.

4.5.1 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and the following. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all requirements specified herein. Test equipment for Government verification inspection shall be made available by the contractor. Inspection of product before preparation for delivery shall consist of groups A, B and C inspection as specified in 4.5.1.3, 4.5.1.4 and 4.5.1.5.

4.5.1.1 Inspection lot. The inspection lot is the quantity of batteries of any one type, of any one code (see 3.10.2) produced at any one place of manufacture on any one contract. At the option of the contractor and at his expense, smaller lots may be selected for inspection.

4.5.1.2 Rejection. If any lot is rejected, the cause for rejection and the corrective action taken in reworking the lot shall be reported in writing to the Government inspector and the contracting officer. Rejected lots shall be resubmitted for reinspection only after all units are reexamined or retested and all defective units are removed or defects corrected. Reexamination information is as follows:

(a) For defects found during group A or B inspection, the Government inspector (and for defects found during group C inspection the contracting officer) shall determine whether normal or tightened inspection shall be used, and whether reinspection shall include all types of defects or only the particular defects which caused initial rejection. If tightened inspection is required, this inspection shall comply with applicable requirements of MIL-STD-105 or, as a minimum, shall use a sampling plan twice the size of the the original sampling plan as required by the Government.

(b) In the event that a test of a lot sample is accidentally terminated due to circumstances which are beyond the contractor's control (e.g., fire, vandalism, power failure) in the opinion of the local Government inspector, the Government inspector shall select a second lot sample to repeat the interrupted test and such subsequent tests as are otherwise required.

4.5.1.3 Group A inspection. This inspection, including sampling, shall conform to Table IV and the inspection procedures of MIL-STD-105, using inspection Level II. Group A inspection shall be performed in the order shown in Table IV.

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TABLE IV - Group A inspection

Inspection	Reg. Para	Insp. Para.	AQL Major	AQL Minor
Visual & mechanical	3.2, 3.4, 3.5, 3.6	4.6	2.5% 1/	6.5%
Internal pressure 2/	3.5.14	4.10.2	1.0%	

1/ Any defect adversely affecting interchangeability and mating of components, performance or life of batteries is considered a major defect.

2/ The internal pressure test may be performed as part of in-process inspections, at the option of the contractor, providing the test is witnessed by the cognizant Government inspector.

4.5.1.4 Group B inspection. This inspection, including sampling, shall conform to Tables V and VI. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. Those batteries to receive the cycle life test of the group C inspection shall be subjected to only the filled discharge capacity and full capacity discharge tests specified in Table V. Type I batteries shall be placed in type II battery boxes or be otherwise restrained during group B testing. The samples subjected to group B inspection, with the exception of samples subjected to group C inspection (see 4.5.1.5), shall not be furnished on contract; they shall be retained by the manufacturer.

TABLE V. Group B inspection

Inspection	Reg. para.	Test para.
Electrolyte Container	3.5.8 (a)	4.10.6
Interchangeability	3.5.7	4.6.1
Filled discharge capacity	3.6.3	4.10.4.3
Full capacity discharge	3.5.12.2	4.10.4.5
High rate discharge	3.5.12.4	4.10.4.6
Extreme temperature	3.6.1.3	4.10.4.7
Insulation resistance	3.5.12.1	4.10.4.1
Immersion	3.6.1.1	4.10.5.4
Vents 1/	3.5.13	4.10.1
Leakage	3.5.12.3	4.10.4.2

1/ The vent tests may be performed as part of in-process inspection at the option of the contractor, provided that the tests are witnessed by the cognizant Government inspector.

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TABLE VI. Sampling plan - group B

Lot size	Sample size	Acceptance number
1-10	0	
11-150	4	0 (Failures)
151-300	6	0 "
301-800	8	0 "
801-1300	12	1 "
1301-3200	16	1 "
3201-8000	24	2 "

4.5.1.5 Group C inspection. This inspection shall be as specified in Table VII and shall normally be performed on 20 percent of the sample units which have been subjected to, and met, group B inspection, with the exception that in no case shall the group C sample lot be less than three batteries. When 20 percent of the sample units results in a fractional unit, one additional battery shall be tested if the fraction is greater than or equal to 0.5. Type I batteries shall be placed in type II battery boxes or be otherwise restrained during group C testing. Group C inspection shall be as follows:

(a) The manufacturer may elect to increase the size of the group C sample lot except that in no case shall the group C sample lot exceed the quantity:

$$2 + \frac{\text{lot size}}{100}$$

(b) For the corrosion resistance test, two samples of each of the replaceable metal piece parts shall be selected in the beginning and the middle of the contract quantity.

(c) Two of the batteries to receive group C tests shall be marked number 1 and number 2 respectively. Batteries number 1 and 2 may be selected from sublots number 1 and 2 at the beginning of each month.

(d) Shipment of a sublot of 100 batteries may be made, at the option of the manufacturer, when the sublot's representative sample has satisfactorily completed its 20th cycle of testing in accordance with 4.11.9, providing that the lot is otherwise acceptable. Each inspection lot must meet all the group C tests including complete cycles of the cycle life test in accordance with 4.10.4.9 before approval for shipment, unless prior approval

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has been granted by the contracting officer. Testing of each sample during the cycle life test must continue until failure and the results reported to the contracting officer. The samples subjected to group C inspection shall not be furnished on contract; they shall be retained by the manufacturer.

TABLE VII. Group C inspection

Inspection	Req. para.	Insp. para.	Sample size
Gassing	3.5.2	4.10.4.10	Battery No. 1 & 2
Retention of charge (Elev. Temp. only)	3.5.12.5	4.10.4.8	Battery No. 1
Vibration	3.6.1(e)	4.10.5.3	Battery No. 2
Cycle life	3.5.12.6	4.10.4.9	All, except No. 1 & No. 2
Corrosion resistance	3.6.2.1	4.10.3.1	See 4.5.1.5 (b)

4.5.1.6 Noncompliance. In addition to the requirements specified in 4.5.1.2, if the Government inspector does not consider that the corrective action will enable the product to meet requirements specified herein or, if the contractor cannot determine the cause for failure, the matter shall be referred to the acquiring activity for resolution. Approval to ship may be withheld, at the discretion of the Government, pending decision of the contracting officer of acquiring activity on the adequacy of corrective action.

4.5.2 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be sampled and inspected in accordance with MIL-P-116 to determine conformance to the requirements specified herein.

4.6 Visual and mechanical inspection. Samples shall be examined to verify that the basic materials, component materials and parts, design and construction, physical dimensions, weight, marking and workmanship are in accordance with all the requirements of this specification and applicable specification sheets. All screws, nuts or bolts shall be torqued as specified by the contractor.

4.6.1 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.5.17:

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

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(c) Location of hinges and fasteners on separate parts of assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings.

4.7 Standard test conditions. Unless otherwise specified herein, all measurements and tests shall be made at a temperature between 19°C to 25°C, ambient atmospheric pressure and relative humidity.

4.8 Test equipment and inspection facilities. When specified (see 6.2), test and measuring equipment, and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.9 Instrument accuracy. Instrument accuracy shall be as follows:

(a) Voltmeters. All batteries subjected to a test load shall be read with a meter having a resistance of at least 1,000 ohms per volt. All voltmeters used shall be accurate within ±0.5 percent of the reading.

(b) Ammeters. All ammeters used shall be accurate within ±1 percent of the full scale reading. The ammeter shall be such that the readings are taken on the upper half of the scale.

(c) Timing equipment. Timing equipment shall be accurate within ±0.5 percent.

(d) Thermometers. Thermometers shall be accurate within 0.5°C.

4.10 Tests.

4.10.1 Vent tests. Vent tests shall consist of the following:

4.10.1.1 Monoblock vent plugs. Monoblock vent plug tests shall consist of the following:

(a) Standard temperature test. The vent shall be removed from the monoblock and attached to an adapter to which a variable pressure source of compressed air is connected. The adapter and attached vent shall then be immersed in water and the air pressure adjusted from the lower limit to the higher limit specified in 3.5.13.1(b). If there is an escape of air bubbles at the lower limit or if there is no escape of air bubbles at the upper limit, the vent has failed the test.

(b) Low temperature test. The above test shall be repeated at -30°C except that the vent shall be conditioned in air at -30°C for 48 hours.

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It shall then be immediately immersed in a container of noncorrosive liquid which has been cooled to -30°C. If there is an escape of air bubbles at the lower limit or if there is no escape of air bubbles at the upper limit the vent has failed the test (see 3.5.13.1(b)).

4.10.1.2 Battery box safety relief valve. The battery box safety relief valve shall be tested in accordance with 4.10.1.1 and meet the requirements of 3.5.13.2.

4.10.2 Internal pressure test. Airtight adapters provided with a valve, gage and controllable air pressure up to 25 lb_f/in^2 (172.4 kPa) shall be inserted in the vent openings of dry charged cells. The cells shall be pressurized at 11 $\pm$1 lb_f/in^2 (75.8 $\pm$ 6.9 kPa) and the valve closed. During a 1-minute stabilization time, the cells shall not show a pressure drop of more than 2 lb_f/in^2 (13.8 kPa). Following the stabilization period, the cells shall not show a pressure drop over a 5-minute period of more than 0.5 lb_f/in^2 (3.4 kPa) from the gage reading at the start of the 5-minute period. Monoblocks will be restrained in a test case during the above test.

4.10.3 Chemical tests.

4.10.3.1 Corrosion resistance. Samples of all external metal parts and the battery assembly container shall comply with 3.6.2.1 after being subjected to the following tests:

(a) Salt spray resistance. The salt spray test shall be conducted in accordance with Method 101 of MIL-STD-202, except that it shall be conducted for a continuous period of 100 hours, using a 5-percent salt solution.

(b) Alkali resistance. Wet one-half of each part of specimen with a solution of 40-percent potassium hydroxide and allow to stand for 48 hours. Specimen shall then be rinsed, dried and observed for the requirements of 3.6.2.1.

4.10.3.2 Alkali resistance of plastic material test. Two specimens, each 2 by 3 inches (50.8 by 76.2 millimeters), shall be cut from each different plastic container or from container materials submitted by the contractor. The cut edges may be sealed with the same resin used in fabrication of the case. After being measured with vernier calipers and weighed in the dry condition at standard test conditions (see 4.7) on a chemical balance, each specimen shall be immersed in a covered vessel containing 40-percent potassium hydroxide. This vessel shall be held for 7 days in an oven at 65°C $\pm$3°C. At the end of the heating period, specimens shall be rinsed in water and dried on the surface. They shall then be inspected for evidence of cracks or blisters, measured and weighed, and the percentage change in dimensions and weight shall be calculated to determine conformance to the requirements of 3.6.2.2.

4.10.3.3 Chemical resistance of elastomer material test. Samples of pads, sleeves, gaskets, O-rings and other items made of elastomer material used in

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the battery shall be immersed for 30 days in 40 percent solution of potassium hydroxide at standard test conditions (see 4.7). They shall then be rinsed with water and allowed to dry. After drying, they shall be inspected for evidence of damage to determine conformance to the requirements of 3.6.2.3.

4.10.3.4 Carbonate determination test. Three randomly selected filled electrolyte containers shall be subjected to this test. Using a volumetric transfer pipette calibrated to deliver 0.5 ml, transfer precisely that amount of electrolyte from a freshly opened electrolyte container sample to a 250-ml Erlenmeyer flask. Add about 25 ml of distilled water and 2 drops of phenolphthalein indicator solution. Titrate with 0.1N(Normality) HCl (Hydrogen Chloride) solution to the end point which will be indicated by a color change from violet to colorless, recording the volume of HCl solution required as V_2. To the colorless sample add one drop of methyl purple indicator. The sample will turn green if it contains potassium carbonate (K_2CO_3) or violet if no potassium carbonate is present. If the sample contains potassium carbonate, titrate it with the 0.1N HCl solution to the end point indicated by a violet color which will occur upon the addition of one or two drops of the HCl solution beyond the appearance of a grey-blue color. Record the total volume of 0.1N HCl solution used for both titrations as V_3. The carbonates content (as K_2CO_3) is calculated as:

$$\frac{2N_1 (V_3 - V_2) \times 69.1}{V_1} \quad \frac{\text{grams}}{\text{liter}}$$

where N_1 is the normality of the HCl solution and
V_1 is the measured volume taken from the electrolyte container.

4.10.4 Electrical tests. Electrical tests shall be performed as follows:

4.10.4.1 Insulation resistance inspection. The insulation resistance shall be measured at a temperature of 22°C ± 3°C and at a relative humidity of 65 ±15 percent, by applying a direct current potential of 500 ±20 volts between any two terminals not electrically connected and between all ungrounded terminals and the case of each activated monoblock and complete battery. The insulation resistance of battery and battery assemblies having nonmetallic cases shall be measured by the use of a 1-inch square (25.4-mm square) copper plate making electrical contact with the case. The plate shall be placed with the broad surface against any area of any surface of the case other than that on which the battery assembly terminals are located. Insulation resistance shall comply with the requirements of 3.5.12.1.

4.10.4.2 Electrical leakage test. The monoblock (removed from the battery) shall be tested for leakage as follows:

(a) The electrolyte in each monoblock shall be adjusted to the manufacturer's recommended operating level.

(b) The outside surface of each monoblock shall be thoroughly washed and scrubbed with water and a brush.

(c) The monoblock shall be dried.

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(d) Invert the monoblock. After 10 minutes, set the monoblock upright and immediately perform the following tests.

(1) Cover to case seal.

a. If the seal between the cover and case is less than one-eighth of an inch (3.2 mm) from top of battery, attach one probe of an ohmmeter to one terminal and the other to a wad of wet cotton. The probe with the wet cotton shall be moved around the seal.

b. If the seal between the cover and the case is one-eighth of an inch (3.2 mm) from top of case, immerse battery in water slightly over the sealing line. Care should be exercised that no water comes in contact with the top of cover or terminals. After immersion for a minimum of 15 seconds, a reading shall be taken on an ohmmeter between the water and one of the terminals.

(2) Terminal to case seal. Attach one probe of ohmmeter to one terminal and the other to a wad of wet cotton. Probe and wet cotton shall be moved around periphery of terminal. (Do not touch terminal.)

(3) Vent plug seal. Conduct test (d)(1)a above, around periphery of vent to cover seal.

(e) A reading on the ohmmeter of less than that specified in 3.5.12.3 shall be considered a defect.

4.10.4.3 Activation and filled discharge. The battery shall be activated according to the manufacturer's instructions. Within 24 hours from the start of filling, the battery shall be discharged at the rate of $0.2C_x$ [4/] to the specified final voltage [5/] or specified ampere hour capacity, whichever occurs first, on the applicable specification sheet. Batteries shall be considered discharged when a specified discharge rate is continued until a specified final voltage is reached or a specified minimum capacity is achieved, whichever occurs first. If the battery fails to meet the specified capacity of the specification sheet, two additional cycles [6/] may be conducted in accordance with 4.10.4.5. If the battery fails to meet the specified capacity by the end of the third discharge, it shall be considered a defect.

4/ The rate of discharge is defined as:

$R = XC_x$

where: R = current rate in amperes
X = multiplying value
C = rated capacity of battery or cell in ampere hours
x = number of hours at which battery rated

For example, $0.2C_1$ indicates a current rate of 5.0 amperes for a 25 ampere-hour battery and that the battery is rated at the 1-hour rate.

5/ The final voltage of the battery is the specified final voltage reached at the end of discharge.

6/ A cycle is a combination of a charge and a discharge.

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4.10.4.4 Charging. All charges shall be conducted under standard condition (see 4.7) and in the upright position. Any charging source which gives the following power may be used:

(a) Constant current:

Charge current: (see applicable specification sheet)
Cutoff voltage: (see applicable specification sheet)

(b) Maximum charging time: (see applicable specification sheet)

Note: Batteries shall be considered fully charged when they meet these requirements.

4.10.4.5 Full capacity discharge. Full capacity discharge procedures shall be as follows:

(a) Batteries shall be charged in accordance with 4.10.4.4, and stored for a minimum of 1 hour at standard conditions (see 4.7) after end of charge.

(b) Discharge battery at $0.05C_x$ at standard conditions for 5 seconds and record voltage. The voltage shall be in accordance with the applicable specification sheet.

(c) Immediately increase current to $0.2C_x$ rate and discharge battery to the specified final voltage or specified ampere hour capacity, whichever occurs first, in the applicable specification sheet.

(d) The battery shall provide the capacity specified on the specification sheet.

4.10.4.6 High rate discharge. High rate discharge procedures shall be as follows:

(a) The batteries shall be charged in accordance with 4.10.4.4 and stored at standard conditions (see 4.7) for a minimum of 30 minutes after end of charge.

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(b) The batteries shall then be discharged at $1C_x$ at standard conditions (see 4.7).

(c) The following shall be recorded:

(1) The terminal voltage 5 seconds after the start of the discharge.

(2) The elapsed time for the voltage to drop to the specified final voltage or to reach the specified ampere hour capacity, whichever occurs first, in the applicable specification sheet.

(d) The battery shall provide the capacity specified on the specification sheet.

4.10.4.7 Extreme temperature tests. Extreme temperature tests shall consist of the following:

(a) Low temperature test. Charge battery in accordance with 4.10.4.4. The battery shall then be stored at the low operating temperature as specified on the applicable specification sheet for 24 hours and discharged at $0.2C_x$ at the same temperature to the specified final voltage or ampere hour capacity, whichever occurs first. The battery shall provide the capacity specified on the specification sheet.

(b) High temperature test. Charge battery in accordance with 4.10.4.4. The battery shall then be stored at the high operating temperature as specified on the applicable specification sheet for 24 hours and discharged at a $0.2C_x$ at the same temperature to the specified final voltage or ampere hour capacity, whichever occurs first. The battery shall provide the capacity specified on the specification sheet.

4.10.4.8 Retention of charge test. Charge battery in accordance with 4.10.4.4. With battery fully charged, it shall be stored at the temperatures and for the periods of time specified on applicable specification sheet. After each storage period, the battery shall be discharged at $0.2C_x$ at the specified storage temperature to the specified final voltage or ampere hour capacity, whichever occurs first. The battery shall provide the capacity specified on the specification sheet. This test shall be repeated for each storage temperature specified on the specification sheet.

4.10.4.9 Cycle life test. Batteries shall meet the requirements of the specification sheet when tested as follows:

(a) On cycles 1-22, charge batteries in accordance with 4.10.4.4. The batteries shall then be discharged without a stand period at $0.25C_x$ for 2.5 hours or to the specified final voltage or to the specified minimum capacity, whichever occurs first.

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(b) On cycles 23-25, the batteries shall be charged at the rate specified in the applicable specification sheet to the cut-off voltage or for the maximum time allowed in the specification sheet for the cycle life test and shall then be discharged at $0.2C_x$ to the specified final voltage or to the specified minimum capacity, whichever occurs first.

(c) After cycle 25, visually check for leakage around terminals and conduct leakage test in accordance with 4.10.4.2. The battery shall meet the requirements of 3.5.12.3.

(d) On cycles 26-50, repeat (a) above until the monoblocks fail to meet the leakage test of 4.10.4.2 or until the ampere hour capacity on any cycle become less than the value shown on the applicable specification sheet, or until the number of cycles completed equals that specified on the specification sheet.

(e) On cycle 51 only, repeat (b) and (c) above.

(f) Over 51 cycles, repeat (d) above until failure.

4.10.4.10 Gassing test. Monoblocks activated with electrolyte shall be fully charged in accordance with 4.10.4.4 under standard test conditions in 4.7. Following this, they shall be placed on OCV stand at 38°C +2°C and shall be held at this temperature for 24 hours. The rate of gas evolution shall be measured at hourly intervals, beginning at the end of the 16th hour after completion of the booster charge and ending at the end of the 24th hour. The rate of gas emission shall not exceed the value specified in 3.5.2.

4.10.5 Environmental tests. Environmental tests shall be performed as follows.

4.10.5.1 Mechanical shock. Mechanical shock tests shall consist of the following:

(a) Test I. Activated with electrolyte, batteries shall be subjected to a total of nine impact shocks, three shocks in each axis (bottom, one side, and one face). The value of each shock shall be 60 gravity units for a duration of 10 to 15 milliseconds. Following this test, batteries shall show no adverse effects when tested in accordance with 3.6.1.

(b) Test II. Type II batteries activated with electrolyte shall be suspended so that one corner is 36 inches (914.4 millimeters) above a 1-inch (25.4-millimeter) white pinewood surface and allowed to fall freely. Repeat test for the other three bottom corners. A very slight abrasion at point of contact is permissible if structural integrity and watertightness of case is not affected. Following this test, the battery shall show no adverse effects when tested in accordance with 3.6.1.

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4.10.5.2 Altitude. Dry and activated batteries shall be placed in a vacuum chamber in the attitude specified in Table II and the pressure reduced evenly, in 12 minutes maximum, to simulate an altitude of 40,000 feet (12.2 kilometers) above sea level. This condition shall be maintained for 1 hour, at which time the batteries shall be inspected to determine compliance with 3.6.1 and the monoblocks tested to determine compliance with 4.10.4.2.

4.10.5.3 Vibration. With battery fully charged, testing shall be performed in accordance with test method 201 (for low frequency conditions) and test method 204 test condition C, part 2 only (for the high frequency conditions) of MIL-STD-202 and the following:

(a) Battery may be vibrated in one direction through both low and high frequencies before being changed to the next direction on the vibration table.

(b) Low frequency vibration may be varied either uniformly or logarithmically between the limits of 10 and 55 hertz.

(c) Length of time for low frequency vibration in one direction shall be 90-100 minutes (270-300 minutes for the complete test).

(d) Voltage shall be monitored on a recording voltmeter during both high and low frequency vibration.

(e) Battery shall be discharged at $0.2C_x$ amperes during the low frequency vibration and the discharge continued to the specified final voltage or the specified minimum capacity, whichever occurs first, on the applicable specification sheet.

(f) At completion of the above test, batteries shall meet the requirements of 3.6.1.

4.10.5.4 Immersion. Type II batteries shall be immersed to a minimum depth of 3 feet (914 mm) in fresh water for 2 hours. After completion of the 2 hour period of immersion, the battery shall be removed and wiped dry on exterior surfaces. When battery box is opened, there shall be no evidence of leakage.

4.10.5.5 Thermal shock. Dry charged batteries, when tested in accordance with test method 107, test condition B of MIL-STD-202 (except that temperatures will be limited to the range -57°C to 65°C) shall meet the requirements of 3.6.1.2(b).

4.10.6 Electrolyte container leak test. The electrolyte containers, filled with the required premeasured quantity of electrolyte and sealed, shall be placed in a test chamber in such position that the level of the electrolyte therein is above the sealed opening of the container i.e., bottle inverted. They shall be subjected two times to the test cycle in Table VIII.

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The rates of temperature and pressure change shall not exceed 1°C per second and 0.5 inch Hg/sec. At the start of each test cycle above, the samples shall be at standard test conditions. After the test, each sample shall be examined for evidence of electrolyte leakage, rupture or incipient failure which might cause electrolyte leakage.

Table VIII. Temperature and pressure profile for electrolyte container leak test

Time (+ 0.1 Hour)	Temperature (+ 3°C)	Absolute Pressure (+1, -2 Inches Hg)
0	-20°C	30
0.5	-20°C	9
1	-40°C	9
1.5	-40°C	30
2	-20°C	30
2.5	0	30
3	30°C	9
3.5	60°C	9
4	60°C	9
4.5	60°C	9

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial as specified in the contract or order (see 6.2).

5.1.1 Level A. Level A preservation and packaging shall be done as follows:

5.1.1.1 Cleaning. Each battery, and auxiliary items, shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each battery, and auxiliary items, shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the following.

5.1.1.4.1 Technical instruction brochure. Each technical instruction brochure shall be packaged method 1C-1 of MIL-P-116.

5.1.1.4.2 Battery. Each battery shall be packaged in accordance with Method III of MIL-P-116. Secure the cover to the battery box with the integral end clamps on the box. Battery shall be wrapped with polyethylene

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plastic material conforming to L-P-378 type I, class 1, grade A and cushioned with a minimum of 1 inch (25.4 mm) of material on all surfaces and cells or pads or both fabricated from fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned battery within a close-fitting box conforming to PPP-B-636, grade W5c, and seal all seams and joints with tape, not less than 2 inches (50.8 mm) wide, conforming to PPP-T-60.

5.1.1.4.3 Monoblock. Each monoblock shall be packaged in accordance with Method III of MIL-P-116. Cushion the monoblock on all surfaces with cells or pads, or both, fabricated from fiberboard conforming to PPP-F-320, type CF, class domestic, variety SW, grade 125, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned monoblock within a close-fitting box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125. Box closure shall be in accordance with the appendix to the box specification.

5.1.1.4.4 Kit, battery filler. Battery filler kits shall be packaged in accordance with Method III of MIL-P-116. Place the required items within a close-fitting box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125. In addition, the kit shall be wrapped with polyethylene plastic sheet and placed in outer fiberboard box conforming to PPP-B-636, grade W5c. Cushion top and sides with 4-1/2 inches (114.3 mm) of vermiculite material and sides with 2 inches (50.8 mm) of vermiculite. Box closure shall be in accordance with the appendix to box specification.

5.1.2 Commercial. Commercial packing shall be in accordance with MIL-STD-1188. This level shall conform to the contractor's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Level A packing shall be done as follows:

5.2.1.1 Battery, storage with battery filler kit (16 bottles). Each battery and one filler kit (16 bottles), packaged as specified in 5.1.1.4.2 and 5.1.1.4.4, shall be packed within a box conforming to PPP-B-601, type overseas, style I. The container with bottles shall be securely separated from the container with the storage battery. Place technical literature, packaged as specified in 5.1.1.4.1, on top of contents, directly under the lid of the box.

5.2.1.2 Monoblock with kit, battery filler (four bottles). Four monoblocks and four battery filler kits (four bottles in each kit), packaged as specified in 5.1.1.4.3 and 5.1.1.4.4, shall be packed within a close-fitting box conforming to PPP-B-636, grade V15c. Box closure shall be in accordance with the appendix to the box specification.

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5.2.2 Level B. Level B packing shall be done as follows:

5.2.2.1 Battery, storage with battery filler kit (16 bottles). Each battery and one filler kit (16 bottles), packaged as specified in 5.1.1.4.2 and 5.1.1.4.5, shall be packed as specified in 5.2.1.1, except that box conforming to PPP-B-601 shall be type domestic, style C.

5.2.2.2 Monoblock with battery filler kit (four bottles). Four monoblocks and four battery filler kits (four bottles in each kit) shall be packed as specified in 5.2.1.2.

5.2.3 Commercial. A quantity of like items, packaged as specified in 5.1.2, shall be packed in a manner to insure carrier acceptance and safe delivery to the first receiving activity.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. In addition to standard markings, interior and exterior containers shall be marked as follows:

"CAUTION - THIS SIDE UP" (WITH ARROWS)

Safety precaution label (corrosive material) in accordance with regulations CFR 49 or Hazardous Material Regulation of the Department of Transportation shall be applied.

6. NOTES

6.1 Intended use. Batteries acquired under this specification and related specification sheets are intended for use where high power output to weight ratio, high discharge rate, flat voltage discharge curve, good shock and vibration resistance, and electrolyte spill resistance are required characteristics. Type I batteries are intended for use as replacement parts for type II or where external battery box protection and restraint will be provided by other means. Type II batteries are intended for use where batteries must contain inherent in-service use protection, such as when used in conjunction with man-pack communication and electronic equipment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification and related specification sheet (see 1.2 and 6.3).

(b) Color of battery box finish (see 3.8).

(c) Battery nomenclature required (see 3.10.1).

(d) When serial numbers shall not be assigned to each battery box cover and monoblock (see 3.10.5).

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(e) Where only type I batteries are acquired, whether the Government or the contractor shall provide type II battery boxes or whether the contractor shall otherwise restrain the type I batteries during testing (see 4.3.2).

(f) When test and inspection facilities shall be established and maintained by the contractor (see 4.8).

(g) Levels of preservation, packaging and packing required (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.	Options
4.3	First Article Inspection Procedure	DI-T-4901	-------
4.3	First Article Inspection Report	DI-T-4902	-------
4.4	Production/Acceptance Inspection Procedures	DI-T-4903	-------
4.4	Production Inspection Reports	DI-T-4904	

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Specification sheet requirements. Since this specification covers only general requirements for parts, materials, processes, design and construction, the specification sheets shall specify actual requirements for the multiple choices or exceptions available in the following paragraphs:

Requirements	Paragraph No.
(a) Classification requirements:	1.2

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(b)	Weight (maximum, filled):	3.7
(c)	Rated ampere-hour capacity ($1C_x$):	3.5.16.1
(d)	Voltage:	3.5.16.2
	(1) Maximum (OCV):	
	(2) Nominal:	
	(3) Final:	
	Down to _____C	
	Below _____C	
(e)	Operating temperature range	3.6.1.2(a)
	(1) High:	4.10.4.7(b)
	(2) Elevated:	4.10.4.8
	(3) Standard:	4.7
	(4) Low:	4.10.4.7(a)
(f)	Activation and filled discharge rate ($0.2C_x$):	4.10.4.3
(g)	Charging requirements:	4.10.4.4
(h)	Full capacity discharge:	4.10.4.5
	(1) 5 sec. voltage (max) ($0.05C_x$):	
	(2) Discharge capacity ($0.2C_x$):	
(i)	Extreme temperature discharge capacity ($0.2C_x$):	4.10.4.7
	(1) Low:_____ A.H.	
	(2) High:____ A.H.	
(j)	High rate discharge	4.10.4.6
	(1) 5 sec. voltage (min) ($1.0C_x$):	
	(2) Discharge capcity ($1.0C_x$):	
(k)	Retention	4.10.4.8
	(1) Elevated temperature storage discharge capacity (28 days):_____A.H.	
	(2) Standard temperature storage discharge capacity (28 days):_____A.H.	
(l)	Cycle life test	4.10.4.9
	(1) Cycles _____ : _____ A.H.	
	(2) Cycles _____ : _____ A.H.	
	:	

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(x) Cycles _____ : _____ A.H.

(m) Drawings required: 2.1.2
(n) National Stock Number: assigned NSN

6.4 First article instructions. Approval of the first articles shall be in writing from the contracting officer. No production units shall be delivered prior to approval of the first articles.

6.5 Standard sample. For information regarding the availability of the standard sample specified in 3.1, address inquiry to the contracting officer.

6.6 Marginal notations. The margins of this specification are not marked with an asterisk to indicate where changes from the previous issued were made due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - MC

Review Activities:
Army - AT, EL

User Activities:
Navy - SH, AS, CG
Air Force - 17

Preparing Activity:
Navy - MC

(Project 6140-0565)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-B-82117D Battery, Storage: Silver-Zinc, Rechargeable; Gen. Spec for

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-LF-014-1802

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